

Believing True Things: Evidence-Grounded Verification for Bengali LLM Hallucination Detection

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Abstract

We describe our system for the অলীকবচন Bengali LLM Hallucination Detection Challenge: given a Bengali prompt, a candidate response, and an optional supporting passage, predict whether the response is faithful or hallucinated. Our system stacks seven per-row signals—lexical grounding, multilingual NLI, two open-weight LLM judges (14B and 32B), self-verification, and retrieval-grounded verification against a Bengali-Wikipedia index—with a per-branch logistic regression trained on the 299 released samples. Our central finding is that the errors of judge-based detectors are *one-sided*: on the no-context split of the validation samples, every row that all baseline signals got wrong was *faithful*. LLM judges do not fail by believing hallucinations; they fail by disbelieving true but obscure, culturally specific facts. Two design responses follow: verdicts with an explicit UNSURE lane that maps to neutrality rather than a wrong “hallucinated,” and retrieval-grounded verification that can prove obscure true answers true. These raised our no-context macro-F1 from 0.694 to 0.778 (out-of-fold) and our public leaderboard score from 0.765 to 0.803. We also report a validation methodology for 299-sample regimes—seed-averaged nested cross-validation—without which we repeatedly mistook fold noise for progress, and we release a fully offline, single-notebook reproduction that recomputes every signal from raw text in under 5 hours on two T4 GPUs.

1 Introduction

Large language models are increasingly the interface through which Bengali speakers reach information, and they are fluent enough in Bengali that their errors read as confident,

natural prose. The অলীকবচন challenge operationalizes this problem as binary classification: each row carries a Bengali prompt, a candidate response, and either a supporting passage (*context* rows) or [NULL] (*no-context* rows); systems predict faithful (1) or hallucinated (0). The benchmark spans cultural-distance bands from globally stable facts (C0) through Bangladesh-specific knowledge (C1) to contested or time-sensitive facts (C2), and only 299 labeled samples are released—a validation set, not a training set.

The two row types are different problems. Context rows reduce to *grounding*: is the response supported by the passage? No-context rows require *closed-book factuality*: is the response true? Our system (§2) treats them as separate branches sharing one stacking recipe. Our contributions are: (i) an error analysis showing that the failure mode of LLM-judge ensembles on culturally specific content is systematic over-skepticism toward true facts (§3); (ii) an evidence-grounded verification branch that directly addresses it (§4); (iii) a validation protocol that survives a 299-sample regime (§5); and (iv) a fully offline reproduction package (§7).

2 System

Every signal is a per-row estimate of $P(\text{faithful}) \in [0, 1]$, with 0.5 meaning “no information.” Signals never make decisions; a per-branch logistic regression stacks them.

Context branch. (1) **substring**: after stripping punctuation and whitespace, is the response a substring of the passage? This trivial rule is remarkably strong (macro-F1 0.894 alone) because most context-row responses are short extractive answers. (2) **nli**: max entailment probability of the response against

sentence-packed passage chunks under multilingual mDeBERTa (Laurer et al., 2024). (3–4) the two LLM judges below, prompted with the passage.

No-context branch. (3) **judge14** and (4) **judge32**: Qwen2.5-14B and Qwen2.5-32B-Instruct (Qwen Team, 2024) (GPTQ Int4 (Frantar et al., 2023)), zero-shot YES/NO factuality judgments. (5) **selfv14** and (6) **selfv32**: self-verification (Manakul et al., 2023)—the model answers the question itself, then a second call judges whether its answer and the candidate agree; the 32B variant uses three-way verdicts with token-level log-probabilities (below). (7) **ret32**: retrieval-grounded verification (§4).

Soft, three-way verdicts. Binary YES/NO parsing discards calibration and, worse, forces a guess when the model has no basis for one. Wherever a signal feeds the 32B model, we prompt for YES/NO/UNSURE, read the first generated token’s top-20 log-probabilities, and set $P(\text{faithful}) = (p_Y + \frac{1}{2}p_U)/(p_Y + p_N + p_U)$. UNSURE thus lands at 0.5—neutral for the stacker—instead of becoming a confidently wrong NO.

Stacking. Signals are stacked per branch by L2-regularized logistic regression refit on the 299 samples at inference time. The decision threshold is the median of per-fold thresholds over 5 seeds $\times$ 5 stratified folds, chosen to maximize macro-F1 on training folds only.

3 Error analysis: skepticism is the failure mode

With six baseline no-context signals (two judges, two self-verifications, a cross-lingual check, and a first-version retrieval branch), we examined the 167 no-context validation rows and asked: which rows does *every* signal get wrong or abstain on? There were 23 such rows, and **all 23 were faithful**. Not one hallucinated row evaded every signal. Meanwhile, inter-judge agreement ranged 0.74–0.86: adding more generic judges adds correlated skepticism, not information.

The picture is specific: when a response asserts a true but obscure fact—a Bangladesh-specific date, a dictionary sense of an id-

iom like ধাক্কা, a grammatical classification—parametric judges of any size default to disbelief. The gap between our system and a perfect one was not “detect more hallucinations”; it was “stop disbelieving true things.” This reframing drove everything that follows: the UNSURE lane (uncertainty must not be spent as a NO vote) and evidence grounding (the only mechanism that can *prove* an obscure true answer true).

4 Evidence-grounded verification

Index. We chunk Bengali Wikipedia (143,069 articles; HF `wikimedia/wikipedia`) into 375,604 passages of $\sim$ 900 characters, up to six per article, embedded with bge-m3 (Chen et al., 2024) (normalized, fp16).

Retrieval. For each no-context row we embed two queries—the question alone, and question + response—take the union of the top-10 neighbors of each, and rerank with bge-reranker-v2-m3 against “*question উত্তর: response*,” keeping the top 4.

Grounded judgment. Qwen2.5-32B receives the evidence passages and the QA pair and must answer YES (evidence supports the answer), NO (evidence contradicts it or shows a different answer), or UNSURE (evidence irrelevant or insufficient), with the instruction to *never* answer NO merely because the evidence does not mention the answer. This matters: verbatim answer-presence in retrieved evidence is non-discriminative (38% of faithful vs. 35% of hallucinated rows), and roughly 40% of evidence sets are irrelevant—notably for word-meaning and grammar questions, where an encyclopedia is the wrong reference. The UNSURE lane routes exactly those cases to neutrality.

Standalone, **ret32** is the strongest no-context signal we measured (macro-F1 0.753 vs. 0.688 for the best judge variant), it recovers 7 of the 23 unanimously missed faithful rows, and the stacker assigns it the largest no-context weight (1.88).

5 Validation in a 299-sample regime

Two early lessons shaped our protocol. First, we once reported an ensemble at 0.8059 out-of-fold; seed-averaging revealed the honest value

System	OOF	no-ctx	LB
substring + constant	0.682	0.345	0.666
+ 14B judge suite	0.795	0.649	0.748
5-signal ensemble	0.793	0.675	0.752
7-signal (two judges)	0.795	0.694	0.765
+ judge32 (binary)	0.803	0.717	0.759
+ selfv32 + ret32	0.833	0.778	0.803
portable 7 (package)	0.831	0.768	0.800

Table 1: Progression. OOF = 5-seed mean macro-F1 on the 299 samples (earlier rows re-scored under the final protocol); no-ctx = no-context branch; LB = public leaderboard. The final row is the fully offline Phase-2 package.

was 0.7946 ± 0.0089 —the gain had been one favorable fold split. Second, a $+0.009$ OOF “improvement” that we submitted *lost* 0.006 on the public leaderboard. Our protocol therefore: (i) every configuration is scored as the mean OOF macro-F1 over 5 seeds of 5-fold stratified CV, with thresholds chosen inside training folds; (ii) differences under ~ 0.02 are treated as noise; (iii) submissions are spent only on gains that clear that bar. Under this protocol, local-to-leaderboard transfer was consistently -0.03 , making the OOF a reliable compass. We also found that a stacker fit on 167 no-context rows saturates quickly: abstention-indicator features, a BanglaBERT classifier fine-tuned on the 299 samples (OOF 0.524, barely above chance—surface form does not reveal factuality), branch-specific feature pruning, and regularization sweeps were all neutral to negative. Signal quality, not stacking sophistication, is the binding constraint.

6 Results

Table 1 traces the system. The two interventions motivated by the error analysis—UNSURE-safe 32B self-verification and evidence-grounded verification—account for the largest single jump ($0.803 \rightarrow 0.833$ OOF, $+0.06$ on the no-context branch), and the public-leaderboard move ($0.765 \rightarrow 0.803$) landed exactly on the -0.03 transfer line. Context-branch performance was near-saturated throughout (~ 0.90 ; substring and the 32B judge are jointly wrong on only 3 of 132 rows), confirming that closed-book verification of culturally specific facts is where this task is decided.

Negative results. Few-shot anti-skepticism prompting produced our best standalone judge (0.688) and rescued 10/23 hopeless rows, yet *hurt* when stacked—with 167 training rows the stacker cannot learn when to trust an aggressive rescuer. Multilingual NLI underperformed the substring rule on context rows (0.635 vs. 0.894): short extractive answers make poor NLI hypotheses. Fine-tuned BanglaBERT was chance-level, as noted above.

7 Reproducibility

Our Phase-2 package is a single Kaggle notebook that recomputes all seven portable signals from raw text—no cached predictions—and refits the stacker at run time. All artifacts (both Qwen GPTQ checkpoints, three encoders, a vLLM wheel set for offline installation, and the wiki index) are pinned as Kaggle artifacts; the notebook runs with internet disabled. Three signals from Phase 1 (a self-consistency 14B judge, a cross-lingual check, and the v1 retrieval branch) required a personal workstation; dropping them costs nothing measurable (OOF 0.831 vs. 0.833), so the package ships the portable seven. Verified end to end on the Phase-1 test set: 4.96 hours on $2 \times T4$ (9-hour budget), 97.7% prediction agreement with our submitted system, public LB 0.800 vs. 0.803. Engineering notes: Bengali runs at roughly one token per character in Qwen’s vocabulary, so grounding prompts need an 8,192-token window; GPTQ was chosen over AWQ because it runs on both P100 (sm60) and T4 (sm75).

8 Limitations and future work

About a third of the remaining unanimously missed rows are word-meaning and grammar questions for which Wikipedia is the wrong corpus; a Bengali Wiktionary index is the obvious next branch. Our wiki snapshot (Nov. 2023) predates recent facts, capping the C2 band. The 299-sample regime caps how much any stacker can learn; our per-branch linear model with seed-averaged thresholds is a floor, not a ceiling. Finally, our analysis is on the released samples; band-level conclusions (C1 vs. C2) await the organizers’ post-competition analysis.

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